H2 ANDERSON JUNIOR COLLEGE
HIGHER 2

CANDIDATE NAME

PDG

PDG INDEX NUMBER

BIOLOGY
9744/02

Paper 2 Structured Questions

12 September 2017
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
Total	100

Answer **all** the questions.

1 Lactose intolerance in humans is the inability to hydrolyse lactose due to the lack of the enzyme lactase in the alimentary canal. As a result, bacteria in the large intestines feed on the lactose and produces fatty acids and methane which lead to diarrhoea and flatulence.

Bacteria have been used to produce lactase on an industrial scale as a dietary supplement for people who are lactose intolerant. Human lactase consists of a single 160 kDa polypeptide chain that localizes to the brush border membrane of intestinal epithelial cells.

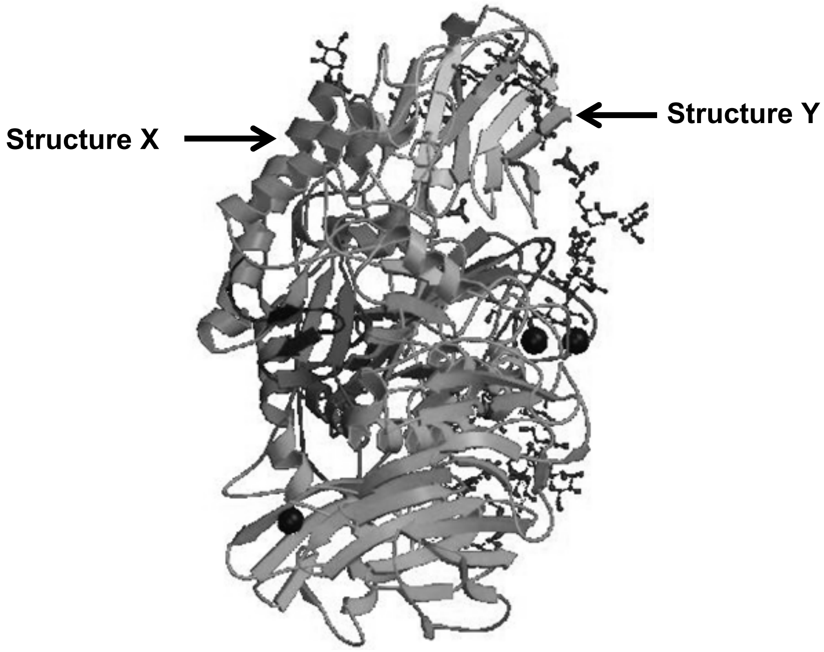


Fig. 1.1

(a) With reference to the **Fig. 1.1**,

(i) state the levels of organization seen in the structure of lactase.

primary, secondary, tertiary structure;

[1]

(ii) Describe structures **X** and **Y**

- structure X- alpha helix, Structure Y- beta pleated sheets
- **repeated coiling/ folding** of polypeptide chain, intrachain hydrogen bond between N-H group of a peptide bond on one fold and C=O of a peptide bond on four amino acids away (alpha helix)/ another fold (beta pleated sheets);

[2]

(b) **Fig. 1.2** shows the hydrolysis of lactose.

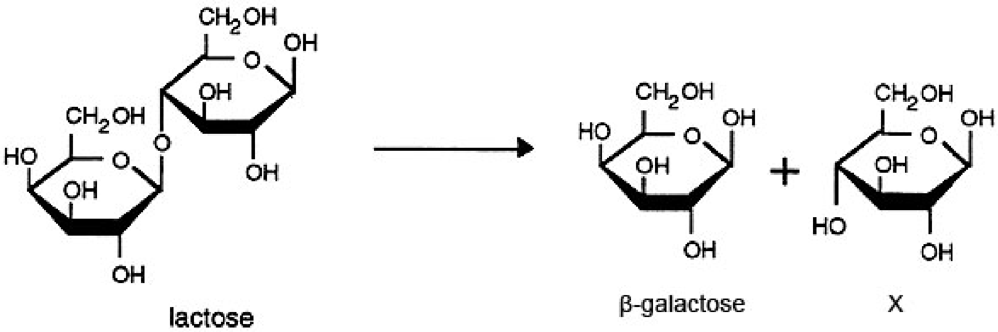


Fig. 1.2

Describe the hydrolysis of lactose, naming the bond that is broken and product **X**.

Use of **water** to break **β-1,4-glycosidic** bond; [2]

to form (either) galactose and **β-glucose**;

(c) Lactase and lactose are protein and carbohydrates respectively.
Explain why there are fewer types of carbohydrate polymers compared to protein polymers.

1. lactose (carbohydrate) is made up of **two** monomers, glucose and galactose but lactase (proteins) are made up of many different types/ 20 types of monomers, amino acids;
2. amino acids differ by their R groups;
3. the different sequences and number of amino acids making up proteins give rise to the large number of different types of proteins / ref to various types of amino acids in a single polypeptide chain vs in a carbohydrate polymer, the same monomer makes up the polymer; [3]

Cell organelles can be separated by centrifuging a cell extract in a sucrose density gradient. The organelles settle at the level in the sucrose solution which has the same density as their own.

The cells used to synthesized lactase were lysed and the cell extract centrifuged in a sucrose density gradient. Three distinct fractions of nuclei, mitochondria and ribosomes (in no particular order) were obtained. The three fractions **A, B** and **C** are shown in the **Fig. 1.3**.

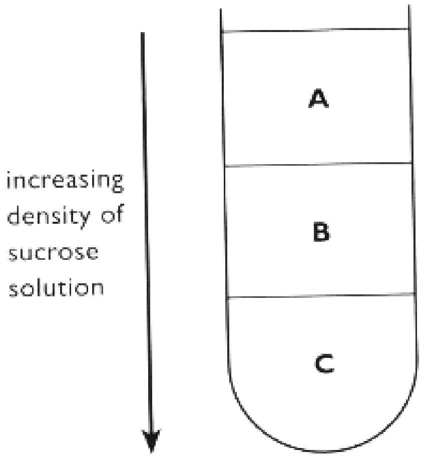


Fig. 1.3.

(d) Identify the organelles in each fraction and describe its role in the synthesis of lactase.

A: Ribosomes – translate genetic message carried by messenger RNA into a polypeptide chain;

B: Mitochondria – sites of cellular respiration, producing energy in the form of ATP which can then be used for peptide bond formation / amino acids activation / exocytosis;

C: Nucleus – contains nucleolus which is responsible for synthesis of ribosomal ribonucleic acids (rRNA) which is a component of ribosomes; OR contains genetic materials (DNA) and the genes found on DNA contains information on how lactase is synthesized;

[1 mark for correct identification of **all** fractions and 1 mark each for each correct function] [4]

[Total: 12 marks]

2 Recently, scientists discovered the presence of a population of bone-marrow derived stem cells that have the ability to form heart muscles cells when transferred to the heart. The stem cells were removed from the bone marrow and cultured so that they divided by mitosis. It was proposed that these stem cells resembled embryonic stem cells.

(a) **(i)** Describe **two** similarities between these bone-marrow derived stem cells and embryonic stem cells.

- **Unspecialised** cell with **no specific structure and function**;
- Ability to **self-renew** via **mitosis**
- Ability to **differentiate** into **specialised** cell types under suitable conditions
- Exhibit **pluripotency**: ability to differentiate into **almost any** cell types except cell of extra-embryonic membranal cells

[2]

(ii) Describe how the rate of mitosis is controlled.

- External growth factors serve as ligands that bind to receptors of cell surface membrane;
- fully- activated receptor activates relay protein which activates a series protein kinases in the phosphorylation cascade;
- Cellular response is the switching on genes that code for transcription factors which will bind to promoter of cyclin genes/ genes that promote or slow down cell cycle;
- Increase transcription and translation of proteins that promote or slow down cell division/ M,S, G1 cyclins/CDKs which control checkpoints in cell cycle;
- **<u>G_1 checkpoint</u>** checks that cell size is adequate/ There is sufficient nutrients are available to support daughter cells/ Growth factors (Extracellular signal proteins that stimulate a cell to grow or divide) are present.
- **<u>G_2 checkpoint</u>** checks that cell size is adequate/ DNA replication is complete and successful/ there is no DNA damage.
- **<u>Metaphase (M) checkpoint</u>** checks that chromosomes are under bipolar tension (in other words, properly attached to kinetochore microtubules originating from the two different poles of the cell)/ chromosomes are aligned at the metaphase plate.
- **Cell cycle checkpoints prevent premature progression of the cell cycle**/ E.g. prevent the segregation of chromosomes before DNA replication is completed.
- **Provides time for cell machinery to be repaired should there be any damage/** E.g. To repair incorrectly replicated DNA sequence.
- Tumour suppressor genes code for proteins that preventing the stimulating activity of cellular proto-oncogenes or oncogenes/ activating DNA repairing genes /activating apoptosis (programmed cell death), preventing uncontrolled cell division.

[4]

(iii) State an advantage of using bone marrow derived stem cells rather than heart stem cells for the treatment of heart diseases.

- Idea of bone marrow stem cells are **easier to isolate / extract** by direct aspiration from the bone marrow in the spinal cord

OR

- Idea of isolation/ extraction of bone marrow stem cells are **less less risky**/ may puncture major blood vessels in removing heart stem cells.

[1]

(c) Troponin is a protein that is integral to muscle contraction in heart muscles. **Fig. 2.1** shows part of its DNA sequence. The entire sequence is 63 base pairs.

5'…GAATTCATGGGCATCGTTGAACAGTGTTGC……………CTTGAGAACTACTGTAACTAAGAATTC…**3'**

3'…CTTAAGTACCCGTAGCAACTTGTCACAACG……………GAACTCTTGATGACATTGATTCTTAAG…**5'**

Fig. 2.1

PCR can be used to confirm presence of troponin DNA sequence. The following pair of primers are used.

Primer	Primer sequence
1	5' AATTCATGGGCATCG 3'
2	5' GAATTCTTAGTTACA 3'

(i) In the boxed area in Fig. 2.1, circle and label the DNA sequences where Primers 1 and 2 will anneal. [1]

(ii) Explain how results of gel electrophoresis of the PCR products are able to show that troponin DNA has been successfully amplified.

- If cloning is successful, a band corresponding to a DNA fragment of 63 bp would be observed;

[1]

(iii) Besides the use of PCR, nucleic acid hybridisation can also be used to determine presence of troponin DNA.

Outline how nucleic acid hybridisation can be used to identify troponin DNA.

- Extract DNA from heart cells, add restriction enzyme and perform gel electrophoresis to separate DNA fragments by size and charge/ Heart cells pressed against a special filter paper;
- which is treated with chemical (NaOH) to burst/lyse the cells and denature the **double-stranded** DNA to obtain **single-stranded** DNA on the filter;
- (Solution containing) chromogenic / labelled / radioactive **single-stranded** nucleic acid/DNA probe complementary to (part of) the troponin DNA is added, probe will hybridise/ bind/anneal to troponin DNA sequence if it is present at areas on the filter paper;
- Carry out/perform autoradiography by placing filter paper on a photographic film; [4]

[Total: 13 marks]

3 The pie chart in Fig. 3.1 shows the relative length of time of each of the stages that ovvur during a particular eukaryotic cell cycle. This complete cycle takes 15 hours.

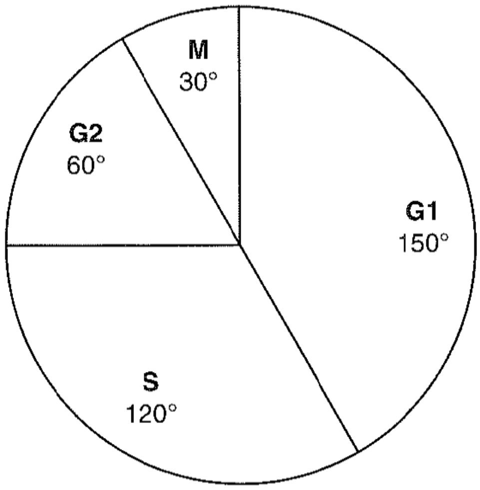


Fig. 3.1

(a) Using the information provided above, calculate how long interphase lasts.

You will lose marks if you do not show your working or do not use appropriate units. [2]

- **Total interphase = 150 + 120 + 60/ 360 – 30 = 330 °**
- **Time in interphase= 330/360 x 15 = 14 hr (to 2 sf)**

(b) During the cell cycle there are a number of checkpoints.

State one function of these checkpoints and explain what might occur as a result of dysregulation of these checkpoints.

Function:

- Ensure that environmental signals (e.g. growth factors) are present for cell division + enough nutrients to support cell division;
- Check that DNA has been replicated without mistakes + DNA repair enzymes to ensure no mutation in cell;
- Ensure there is bipolar tension for every chromosomes so that sister chromatids can separate to opposite pole/ maintain chromosome number;

(any 1)

Dysregulation:

- Cell divides even without growth factor/ mutation during DNA not detected or repaired and passed on to daughter cells/ sister chromatids not separated to opposite poles leading to changes in chromosome number or gain of function mutation of protooncogene to oncogene;
- Uncontrolled cell division; [3]

(c) Colorectal cancer is one of the most common cancers in Singapore. Cancer of the colon and rectum – colorectal cancer – begin as polyps (also known as adenoma) that grow on the inner lining of the large intestine.

Most sporadic cases of colorectal cancer are believed to develop from benign adenomas (polyps) to carcinoma by the accumulation of genetic abnormalities as shown in Fig. 3.2.

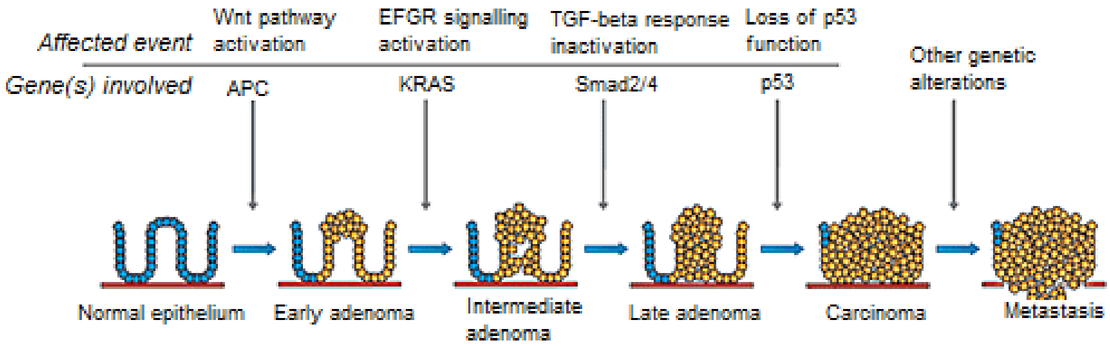


Fig 3.2

(i) Using the Figure 3.2, explain why the development of cancer as a multi-step process.

- Requires accumulation of mutation in a single cell/ same lineage;
- Mutation of APC activates Wnt signaling pathway that leads to uncontrolled cell division, forming early adenoma;
- Accumulation of mutation in KRAS activates EGFR signaling pathway induces cell to divide even without growth factor/ Smad2/4 inactivates TGFβ response which causes increase cell proliferation/ cell division is uncontrolled and excessive;
- Loss of function mutation of tumor suppressor gene p53 results in formation of a carcinoma;
- Further genetic mutations causes metastasis; [5]

(ii) The majority of all colorectal cancers occur sporadically without any known cause, but certain groups of people have a predisposition to the development of cancer of the large intestine. These people may carry specific genetic mutations or have relatives with the condition.

Approximately 15% of all colorectal cancer cases are familial, with the most common inherited conditions being familial adenomatous polyposis (FAP). Patients with FAP have a lifetime risk of the development of colon cancer that approaches 100%. Patients with FAP have a germline inactivation of one APC allele. Adenoma formation is faster, but progression from adenoma to carcinoma has the same rate as sporadic colorectal cancer as shown in Fig. 3.3.

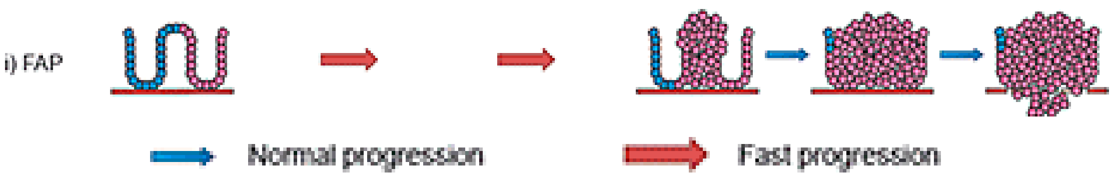


Fig. 3.3

Using the information above and your own understanding of the development of cancer, suggest why patients with FAP form adenoma faster.

- FAP patients only need to have one more copy of their APC allele to undergo a loss-of-function mutation;
- Ref. to APC as tumor suppressor gene;
- Need to accumulate less mutations for adenoma formation and therefore formation is faster;

[2]

4 Genome editing is the process in which a DNA target sequence is replaced by a desired sequence. **Fig. 4.1** shows how the process is being done by Cas9 enzyme which makes use of a guide RNA to achieve the editing effect. This can be done in embryos so that those children who are born of parents with the genetic disease alleles would not suffer from the genetic dsease.

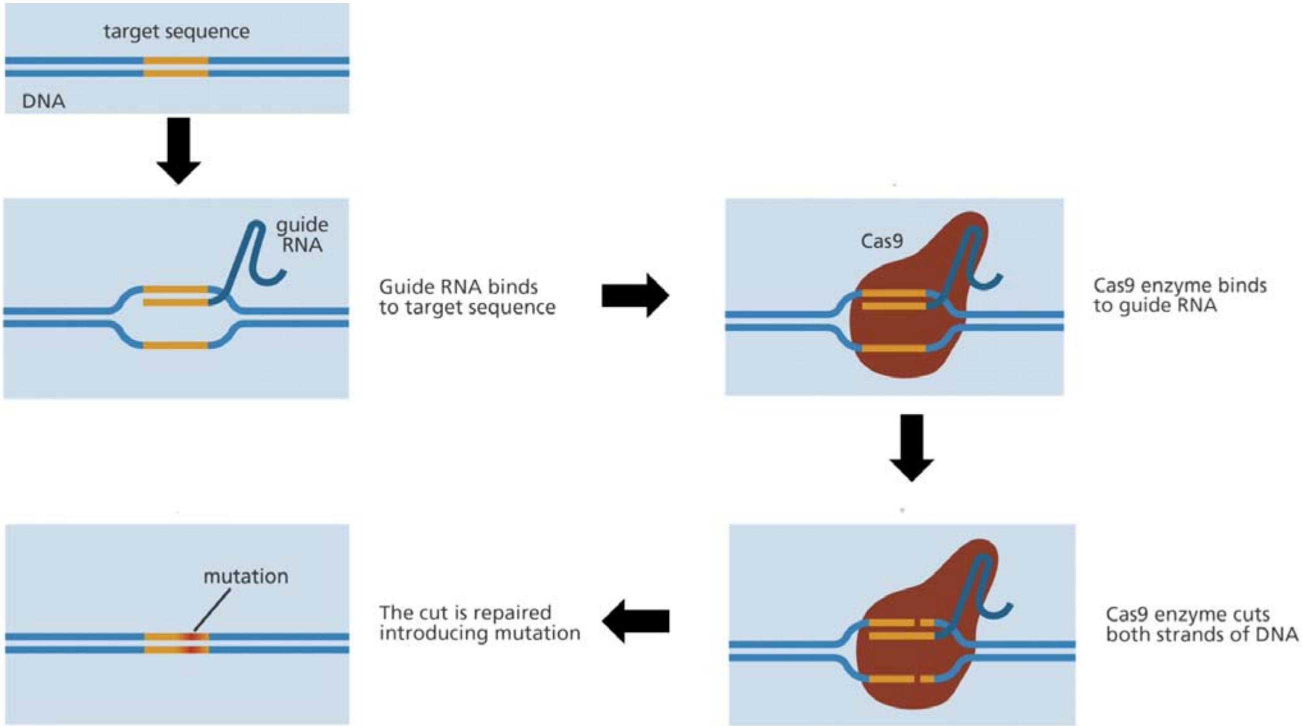


Fig. 4.1

(a) With reference to Fig. 4.1, describe how the guide RNA and Cas9 enzyme are used to cut both strands of DNA. [3]

- Hydrogen bonds form between single-stranded guide RNA with complementary bases to 1 target DNA strand;
- Cas9 has active site which binds to guide RNA;
- Cleaves both strands of DNA by breaking phosphodiester bonds;
- **Additional marking point:** Active site of Cas9 is complementary in shape to target base sequence;

(b) **(i)** Explain why gene editing done on embryos help to prevent children born from suffering from the genetic disease. [2]

- All organs and tissues are derived from the embryos by mitosis and differentiation;
- All daughter cells are genetically identical to the edited embryos;
- **Additional marking point:** Embryonic stem cells are pluripotent;

(ii) Suggest and explain if the mutation introduced by gene editing as shown in Fig. 4.1 should be dominant or recessive. [2]

- Dominant mutation;
- Codes for a functional protein to mask the effect of recessive allele;

(c) Describe how mutations in DNA arise in nature. [2]

- A named factor (e.g.UV light, ionizing radiation, tar in cigarette smoke) as cause;
- Proof-reading by DNA polymerase or DNA repair mechanisms did not correct mutation;
- Any valid point

RNA plays a very important role in many biological processes. One of them is transfer RNA (tRNA) which has extensive intramolecular hydrogen bonds.

(d) **(i)** State **two** importance of having such bonds. [2]

- Confers stability;
- Confers a 3D shape;

(ii) Relate the structure of tRNA to its functions. [2]

- Has 3' CCA end to form covalent bond with amino acid;
- Its 3D shape allows it to fit into the active site of amino-acyl tRNA synthetase;
- Has anticodon which forms complementary base pair with mRNA codon;

(Any 2)

[Total: 13 marks]

5 Figure 5.1 represents a bacteria DNA and a eukaryotic chromosome in metaphase of mitosis, not drawn to scale.

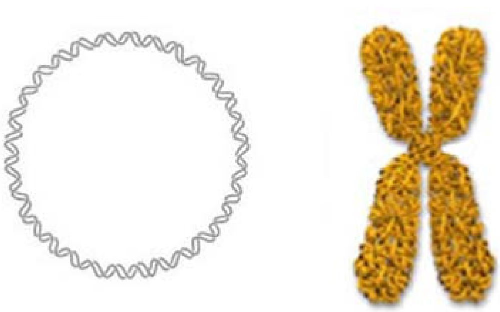


Fig. 5.1

(a) State **two** ways in which the organization of genes found in these two structures differ and suggest **one** advantage of this to the bacterium.

Feature	Eukaryotic	Prokaryotic
Gene organization	Monocistronic genes	Polycistronic genes / **operons**
Advantage to bacteria	Simultaneous expression of closely-related genes organised in an operon	
Association between DNA and histones	Association with **histones** / scaffolding	No association with histones

	Proteins - allows for increased structural complexity/folding to higher degree of condensation e.g. between euchromatin and heterochromatin states	- does not achieve same level of condensation complexity as eukaryote

[3]

(b) In 1946, Joshua Lederberg and Edward Tatum proposed that bacterial cells undergo genetic recombination. To test their hypothesis, they experiments using two bacteria strains of *Escherichia coli (E.Coli*) with different nutritional requirements.

Strain A, B and a mixture of both strains were grown on culture plates containing minimal medium that does not contain essential amino acids. The results are shown in Fig. 5.2.

Mutant genes (–) do not code for enzymes that synthesize amino acids. Note that all five amino acids are required for bacterial growth.

Bacterial strains	Genes for biosynthesis of amino acids	Mutant genes for biosynthesis of amino acids
A	thr^+ leu^+ thi^+	met^- bio^-
B	met^+ bio^+	thr^- leu^- thi^-

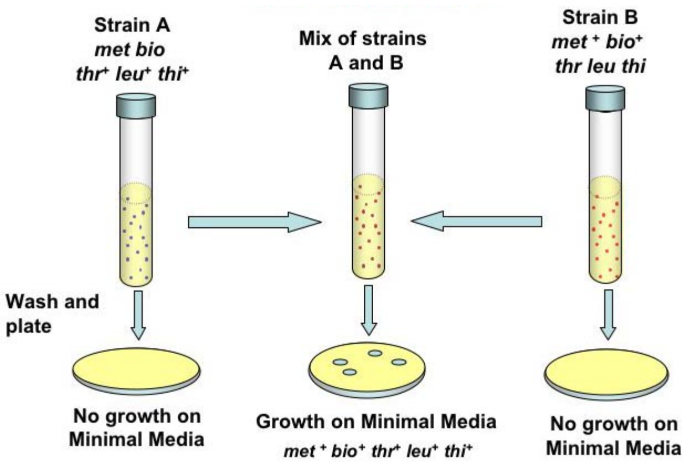


Fig. 5.2

Another researcher, Bernard Davis also worked with the same hypothesis. In his experiment he constructed a U-tube in which the two arms were separated by a fine filter. The pores of the filter were too small to allow bacteria to pass through but large enough to allow easy passage of the fluid medium, any dissolved substances and free DNA. The results are shown in Fig. 5.3.

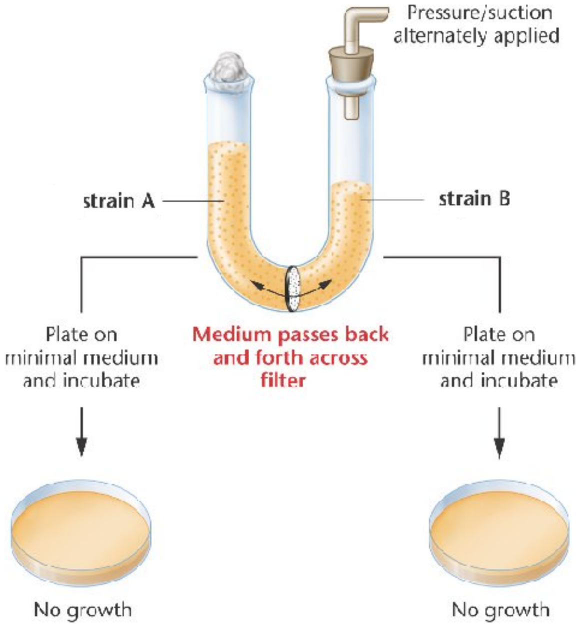


Fig. 5.3

(i) Using the results of the two experiments and your understanding of genetic recombination in bacteria, state the genetic recombination that has taken place between Strain A and B. Explain your answer.

- Conjugation;
- Second experiment shows transduction and transformation did not take place;
- Because no colonies grew on minimal medium agar;
- genetic recombination occurs only when physical contact is possible between 2 strains;
- In first experiment, bacteria that grew on minimal medium has DNA that encodes for all essential amino acids/ ref. recombinant bacteria has grown on minimal medium
- Showing the genes from Strain A have transferred to Strain B / converse through formation of conjugation tube/ ref. to conjugation tube [6]

(c) In 2016, a pathogenic strain of *E.Coli* found on unwashed salad caused food poisoning in 151 people in Britain, leaving two of them dead.

Using a named example, describe how such pathogens are usually treated. [3]

- Antibiotics
- Penicillin
- Competitive inhibitor to transpeptidases, prevent cell wall synthesis;
- Resulting in bacterial cell lysis;

[Total: 12 m]

6 In anaerobic respiration in yeast, the pyruvate molecules are broken down to produce ethanol and carbon dioxide. The release of carbon dioxide can be used to investigate the rate of anaerobic respiration.

Fig. 6.1 shows and experiment which was set up to find the rate of anaerobic respiration.

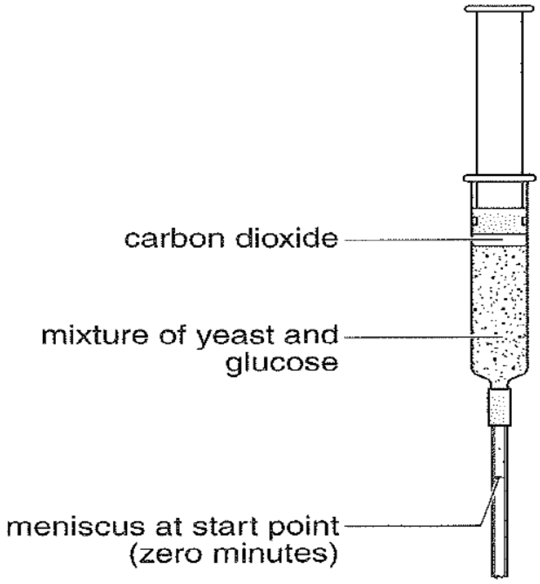


Fig. 6.1

The meniscus moves down the tube as carbon dioxide is released.

Table 6.1 shows the distance moved by the meniscus from the start point. This was recorded every 10 minutes.

Table 6.1

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	1	2	5	9	14	21	45	73	98

(a) The rate of anaerobic respiration can be calculated by using the rate of movement of the meniscus.

Calculate the rate of anaerobic respiration between 70 and 80 minutes.

You will lose marks if you do not show your working.

Rate of Anaerobic Respiration = (73-45) mm / (80-70) min
= 28 mm / 10 min
= **2.8** mm min^{-1}

[2]

(b) This experiment was repeated three more times. Each time, the glucose (a monosaccharide) was replaced with a different disaccharide sugar:

- Maltose – a disaccharide of glucose and glucose
- Sucrose – a disaccharide of glucose and fructose
- Lactose – a disaccharide of glucose and galactose.

Tables 6.2 (a), (b) and (c) show the results of these experiments.

Table 6.2 (a): Using maltose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	0	0	2	3	6	9	12

Table 6.2 (b): Using sucrose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	1	3	11	22	37	48	61

Table 6.2 (c): Using lactose

Time/ min	0	10	20	30	40	50	60	70	80	90
Distance travelled by meniscus from start point/ mm	0	0	0	0	0	0	0	0	0	0

With reference to the information provided in Tables 6.2 (a), (b) and (c) and your biological knowledge:

(i) Describe the difference in the results for maltose and sucrose, and suggest one explanation for this difference,

- **Glucose** is the respiratory substrate for **glycolysis**;
- Maltose: meniscus only starts moving at 50 min vs sucrose at 30 min/ movement of sucrose more than maltose + data;
- Enzyme to break down sucrose is more readily available/at higher concentration than enzymes to break down maltose; [2]

(ii) Suggest two explanations for the results for lactose.

- Yeast does not have proteins channels that allows uptake of lacotse;
- Yeast does not encode for lactase that breaks down lactose to glucose and galactose;
- AVP; [2]

(c) An electron micrograph of yeast, *Candida albicans*, is shown in Fig. 6.2.

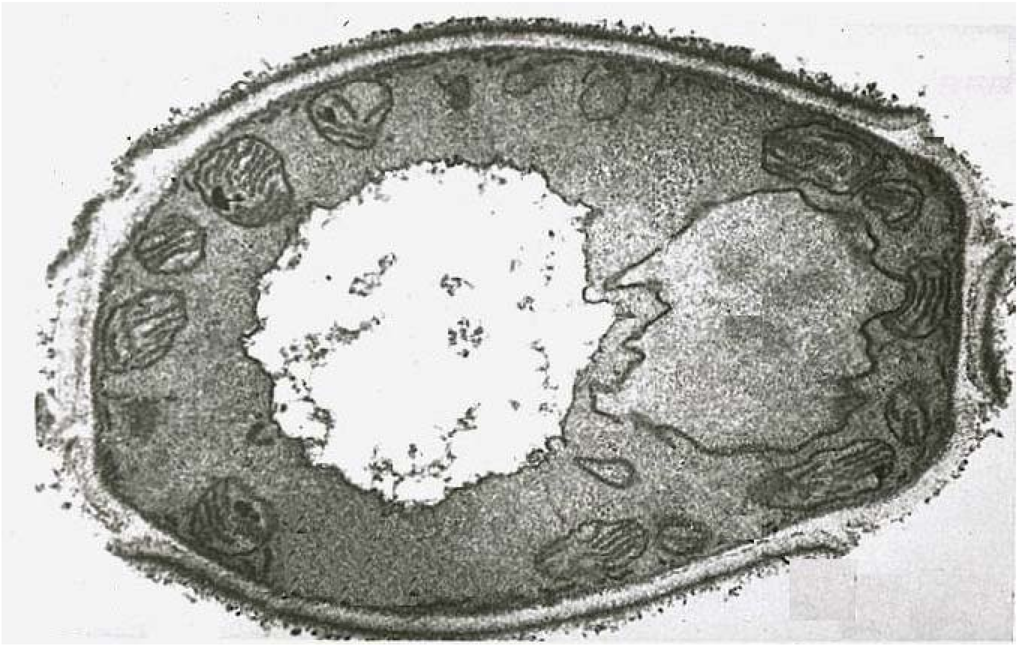


Fig. 6.2

(i) On Fig. 6.2, label site of

i. Glycolysis

ii. Oxidative phosphorylation [2]

(ii) State one visible structure of mitochondria from Fig. 6.2 and describe how it supports mitochondria's function.

- Highly folded inner membrane + Increase surface area for embedment of more ETC/ ATP synthase to increase rate of ATP production
- Membrane bound/ double membrane + Enclose matrix that contains enzymes for link reaction and Kreb's cycle/ compartmentalizes matrix for optimum conditions for enzymes to work; [1]

(ii) Besides location, compare between oxidative phosphorylation and photophosphorylation. [4]

Features	Oxidative phosphorylation	Photophosphorylation
Functions in the presence of..	oxygen	light
Source of energy	NADH and FADH2	light
No. of electron transport chain	1	2
Electron flow	linear – one-way	linear or cyclic
Final electron acceptor	oxygen	NADP (non-cyclic) PSI reaction center (cyclic)
Involvement of water	water produced	photolysis of water
Establishment of proton gradient	protons pumped <u>outwards</u> from <u>matrix</u> across <u>inner mitochondrial membrane</u> into <u>intermembrane space</u>	protons pumped <u>inwards</u> from <u>stroma</u> across <u>thylakoid membrane</u> into <u>thylakoid space</u>
Products	ATP, water	ATP, NADPH, oxygen

[Total: 13 marks]

7 Pigment production in onions is controlled by two enzymes resulting in three different coloured bulbs - red, yellow and white.

A pure-white strain crossed with a pure-red strain produces an all-white F_1. Two F_1 onions with white bulbs were crossed. The F_2 generation was found to consist of 2170 white, 530 red and 180 yellow-bulbs.

The alleles are represented by the following symbols:

I: no production of pigment | **i**: production of pigment
R: red pigment | **r**: yellow pigment

(a) State the mode of inheritance in the onion bulb colour. [1]

- (Dominant) epistasis

(b) Explain the results of the cross by drawing a genetic diagram in the space below. [5]

Parental phenotypes: **White bulb** **X** **Red bulb**

Parental genotypes: **IIrr** **X** **iiRR**

Gametes: **Ir** **X** **iR** **[1]**

F_1 phenotypes: **White bulb** **X** **White bulb**

F_1 genotypes: **IiRr** **X** **IiRr** **[1]**

Gametes: **IR** **Ir** **iR** **ir** | **IR** **Ir** **iR** **ir** **[1]**

Punnett Square:

	IR	Ir	iR	ir
IR	**IIRR**	**IIRr**	**IiRR**	**IiRr**
Ir	**IIRr**	**IIrr**	**IiRr**	**Iirr**
iR	**IiRR**	**IiRr**	**iiRR**	**iiRr**
ir	**IiRr**	**Iirr**	**iiRr**	**iirr**

F_2 genotype ratio: **12 I_ _ _** **:** **3 iiR_** **:** **1iirr** **[1]**

F_2 phenotype ratio: **12 White** **:** **3 Red** **:** **1 Yellow** **[1]**

(c) Explain how the different phenotypes results. [4]

- Allele I is dominant over the R/r locus;
- I codes for an inhibitor as a first step;
- In absence of inhibitor allows R allele codes for an enzyme that results in red pigment;
- In absence of inhibitor allows rr genotype results in a different enzyme that results in yellow intermediate;

(e) Suggest how a farmer may determine if a red onion is homozygous in both loci. [2]

- Cross a red onion with a yellow onion (iirr);
- If it is homozygous in both loci, only red onion offsprings will result;

[Total: 12 marks]

8 Antibodies against tuberculosis are produced by plasma cells during an immune response.

Fig. 8.1 shows a diagram of an antibody molecule.

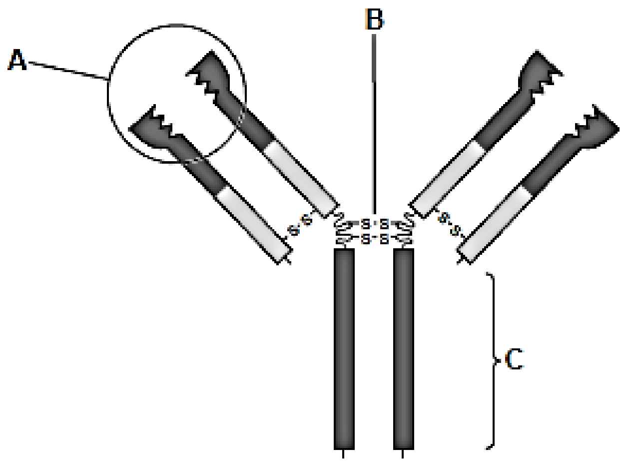


Fig. 2.1

(a) Explain the functions of the parts labelled A, B and C.

(i) A

Variable region forms the antigen-binding sites;
Binds/attaches/combines, to antigen;
Each antigen-binding sites (has a specific 3D shape) is specific to the shape of one antigen. [2]

(ii) B

Disulphide bond holds polypeptides/heavy chains together;
Maintains tertiary/quarternary/3D shape/structure;
The 'hinge' region gives the flexibility for the antibody molecule to bind around the antigen; [1]

(iii) C

Constant region binds to receptors/ cell surface membrane on phagocytes/ macrophages;
Antigen marking/tagging for phagocytosis/macrophage action;
Opsonisation - Many phagocytic cells bear receptors for the Fc portion of the antibody and adhere to the antibody-coated bacteria, leading to engulfing and destruction of the microorganism. [1]
Agglutination/ precipitation of antigen - Insoluble antigen-antibody complexes are easily phagocytized and destroyed by phagocytic cells

(b) Explain why tuberculosis (TB) is known as an infectious disease. [3]

1. Caused by a pathogen, bacterium, mycobacterium tuberculosis;
2. Transmits easily from infected individual to other uninfected individual via various mode of transmission - aerosol/airborne droplet from infected person exhaling/coughing/sneezing/shout/sing;
3. **Depending on the environment**, these tiny particles can remain **suspended in the air for several hours**;
4. Transmission occur when a person **inhales** and the droplet nuclei transverse the mouth or nasal passages, upper respiratory tract, and bronchi to reach the alveoli of the lungs.
5. Person drinks unpasteurized milk/ eats meat from infected cattle;

(c) Outline the role of antibiotics in the treatment of infectious diseases, such as TB. [3]

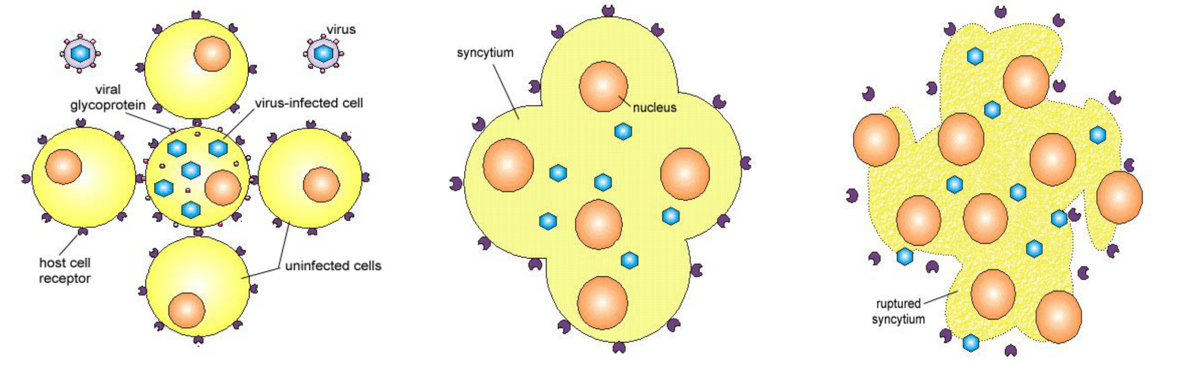
1. Kill bacteria/bactericidal/ cause bacteria to lyse/ swell and burst;
2. (or) bacteriostatic/prevents bacterial growth/prevents bacterial replication;
3. Antibiotics interferes with the modulation of chromosomal supercoiling through topoisomerase-catalyzes strand breakage and rejoining reactions is required for DNA synthesis, mRNA transcription and cell division.
4. Prevents protein synthesis (Initiation and elongation)/ inhibit RNA polymerase;
5. Antibodies may also result in protein mistranslation by promoting tRNA mismatch with mRNA codon.
6. Antibiotics may also Inhibit cell membrane function which result in leakage of important solutes essential for the cell's survival.
7. Prevent spread of bacteria within body/ prevents formation of pathogen reservoir for re-infection;
8. Do not affect human cells/ tissues/ not toxic to humans;
9. Prevents death/ consequences may be fatal if no antibiotic treatment/ alleviate symptoms/ faster recovery;
10. Prevent transmission/ spread of disease (do not confuse with mp 4);

(d) While TB is a bacterial infectious disease, HIV is a viral infectious disease.

Explain how HIV cause diseases in humans through the disruption of host tissue and functions.

HIV infection causes destruction of T helper cells by the following mechanisms: [3]

1. Hijacking of cellular machinery and resources towards producing new virus disrupts normal activities needed for cell survival, eventually causing **cell death**.
2. Budding of a large amount of viruses from the cell surface might also disrupt the cell membrane sufficiently for host cell to die.
3. HIV may induce adjacent T helper cells to fuse together forming giant multinucleated cells or **syncytia**.
4. Shortly after syncytia formation occurs, the fused cells lose immune function and die.
5. functional T helper cell levels decline to a critical point



Uninfected cells bind to a virus infected cell.

The uninfected and infected cells fuse forming a multinucleated giant cell or syncytium.

The syncytium eventually becomes too large and ruptures.

Fig 36: Formation of a syncytium *(singular).*

[Total: 13 marks]